

Un ideal es maximal si y solo si el cociente es un cuerpo

Proposición 1. Sea $I \subsetneq R$ un ideal, entonces

$$I \text{ es un ideal maximal} \iff R/I \text{ es un cuerpo.}$$

Demostración: Podemos ver la equivalencia directamente:

$$I \text{ es maximal} \iff I \neq R \wedge \left(\forall J \subset R \text{ ideal} : I \subset J \subset R \implies I = J \vee J = R \right)$$

Por el teo-correspondencia-ideales-cociente/Teorema 1, $\forall J \subset R$ ideal que contiene a I , existe un único ideal $\bar{J} \subset R/I$ tal que $\pi^{-1}(\bar{J}) = J$, donde $\pi : R \rightarrow R/I$ es el morfismo canónico. Así,

$$I \text{ es maximal} \iff R/I \neq \{0\} \wedge \left(\forall \bar{J} \subset R/I \text{ ideal} : \bar{J} = \{0\} \vee \bar{J} = R/I \right).$$

Por el lem-cuerpo-iff-ideales-triviales/??, esto es equivalente a que R/I sea un cuerpo. ■

Referenciado en

- teo-extension-entera-ideal-primo-maximal-iff-maximal
- cor-ideal-maximal-imp-primo

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